Determine the derivative of:

$$f(x) = \frac{\sin x}{1-\cos x}$$

$u = \sin x$ $\quad$ $u' = \cos x$

$v = 1-\cos x$ $\quad$ $v' = \sin x$

$$f'(x) = \frac{\cos x(1-\cos x) - \sin x \sin x}{(1-\cos x)^2}$$

Simplify:

Let $L = \cos(1-\cos x) - \sin x \sin x$

$L = \cos x - \cos^2 x - \sin^2 x$

We know: $\sin^2 x + \cos^2 x = 1$

So: $-\cos^2 x - \sin^2 x = -1$

$\Rightarrow L = \cos x - 1$

$$f'(x) = \frac{\cos x - 1}{(\cos x - 1)^2} = \frac{1}{\cos x - 1}$$

Determine the derivative of:

$$g(x) = (2+\ln x)(2-\ln x)$$

$$u = 2+\ln x \qquad u' = \frac{1}{x}$$

$$v = 2-\ln x \qquad v' = -\frac{1}{x}$$

$$g'(x) = \left(\frac{1}{x}\right)(2-\ln x) + (2+\ln x)\left(-\frac{1}{x}\right)$$

Simplify,

$$g'(x) = \frac{2-\ln x}{x} + \frac{-2-\ln x}{x}$$

$$\boxed{g'(x) = \frac{-2\ln x}{x}}$$

Prove that :

$$\int x\cos x \; x\,dx = x\sin x + \cos x + C$$

$$\frac{d}{dx}\left(x\sin x + \cos x\right) = \sin x + x\cos x - \sin x$$

$$\Rightarrow \boxed{x\cos x}$$

Proved!